

The first  $(\log n)^2$ : the three-dimensional standard integral

$$Z_3(n) \sim \frac{1}{16} S_{1/2}(a) n^{-1/2} (\log n)^2$$

Claude, with Daniel Murfet

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### Abstract

For  $d = 3$ ,  $k = (1, 1, 1)$ ,  $h = (0, 0, 0)$  all three candidate exponents equal  $\frac{1}{2}$ , so the multiplicity is  $m = 3$  and the paper's expansion carries  $(\log n)^{m-1} = (\log n)^2$ . Iterating the two-dimensional reduction once more,

$$Z_3(n) = \int_0^b \int_0^b \int_0^b e^{-\beta n(uvw)^2 + \beta \sqrt{n} uvw a} = n^{-1/2} H_2(\sqrt{n} b^3), \quad H_2(L) = \int_0^L \frac{H(x)}{x} dx,$$

and integrating the unit-35 squeeze  $H(x) = A \log x + O(1)$  against  $dx/x$  gives  $H_2(L) = \frac{A}{2} \log^2 L + O(\log L)$ , hence  $Z_3(n) \sim \frac{A}{8} n^{-1/2} \log^2 n = \frac{S_{1/2}(a)}{16} n^{-1/2} (\log n)^2$ . Zero `sorry/axiom`.

## 1 Setting

The  $\log n$  powers in `thm:TaylorTree` count coincidences among the candidate exponents. Unit 35 gave  $m = 2$ ; the natural next question is whether the same one-dimensional machinery produces  $m = 3$ . It does: the inner two integrals reduce (by `twoD_inner` and the “divide and substitute” lemma  $\int_0^b F(cu)/u du = H(cb)$ ) to  $n^{-1/2} \int_0^b H(\sqrt{n} b^2 u)/u du$ , and the same lemma for  $H$  gives  $H_2(\sqrt{n} b^3)$ . Each extra dimension with a coinciding exponent adds one more  $\int dx/x$ , i.e. one more power of  $\log$ .

## 2 The things formalised

```
theorem logLogPrimitive_bounds ( a L : ) (h : 0 < ) (hL : 1 ≤ L) :
  A * Real.log L ^ 2 / 2 - M * Real.log L = logLogPrimitive a L
  logLogPrimitive a L ≤ A * Real.log L ^ 2 / 2 + E * Real.log L + E
  -- (A = quadMass a, M = quadMoment a, E = exp ( a^2/2))
theorem threeDIntegral_eq ... :
  threeDIntegral a b n = (√n) ^ 1 * logLogPrimitive a (√n * b ^ 3)
theorem threeDIntegral_isEquivalent ( a b : ) (h : 0 < ) (hb : 0 < b) :
  (fun n => threeDIntegral a b n) ~[atTop]
  fun n => fluctuation (1/2) a / 16 * (n ^ (-(1/2 : )) * Real.log n ^ 2)
```

Supporting lemmas:  $|F(t)| \leq E|t|$  on all of  $\mathbb{R}$  (so  $F(t)/t$  is bounded and  $H$  has a continuous interval-integral representative),  $0 \leq H(x) \leq Ex$ , integrability of  $H(x)/x$ ,  $\int_1^L \log x/x = \frac{1}{2} \log^2 L$  (FTC with  $(\log x)^2/2$ ), the substitution lemmas `integral_quadPrimitive_div_eq` and `integral_logPrimitive_div_eq`, and an isolated algebraic helper `logsq_bound_helper` turning the  $H_2$  squeeze at  $\log L = \frac{1}{2} \log n + 3 \log b$  into  $|H_2 - \frac{A}{8} \log^2 n| \leq K(\log n + 1)$ .

### 3 Proof strategy

Exactly the unit-35 pattern one level up. For the squeeze, split  $(0, L]$  at 1: on  $(0, 1]$  the integrand  $H/x \leq E$ ; on  $(1, L]$  sandwich  $H(x)/x$  between  $(A \log x \mp \text{const})/x$  using `logPrimitive_bounds`, and integrate exactly. The three substitutions are `intervalIntegral.integral_comp_mul_left` with the everywhere-valid identity  $G(cu)/u = c \cdot G(cu)/(cu)$  ( $G = F$  or  $H$ , both vanishing at 0). The final  $o$ -comparison is  $(\log n + 1)/\sqrt{n} = o(\log^2 n/\sqrt{n})$ , i.e.  $1/\log n + 1/\log^2 n \rightarrow 0$ .

### 4 Roadblocks and resolutions

- Membership in `Ioc 1 L` gives  $1 < x$ , not  $1 \leq x$ : use `one_pos.trans hx.1` for positivity, `hx.1.le` for the bounds.
- A direct `nlinarith` on the  $(\log n)^2$  bound (with  $\log L$  substituted) failed; isolating the pure algebra in `logsq_bound_helper` with the needed product nonnegativities named made it a routine `nlinarith`, and the main proof applies it by `convert ...using 2; ring`.
- `private` lemmas of an earlier file (the bounded-domination integrability of  $1/x, 1/x^2$ ) are not reusable; here the same facts came from continuity on  $[1, L]$  instead.
- The unweighted `H` has no interval-integral definition; give it one for  $x \geq 0$  (`logPrimitive_eq_intervalIntegral`) and take measurability from the continuous primitive of the bounded integrand  $F(t)/t$ .

### 5 What was Mathlib, what was new

Mathlib: `intervalIntegral.continuous_primitive`, `integral_eq_sub_of_hasDerivAt`, `Real.hasDerivAt_log`, `integral_inv_of_pos`, `integral_comp_mul_left`, `tendsto_log_atTop`.  
New: the iterated logarithmic primitive and its squeeze, the three-dimensional reduction, and the first formal  $(\log n)^2$  asymptotic.

### 6 Lessons

The multiplicity mechanism is now completely transparent in Lean: each coinciding candidate exponent contributes one “divide by  $u$  and substitute” step, i.e. one more  $\int dx/x$  and one more power of  $\log n$ ; the analytic content is a single uniform  $O(1)$  squeeze propagated through these steps.

### 7 Follow-ups

General  $d$  by induction on the iterated logarithmic primitive ( $H_m(L) = \frac{A}{m!} \log^m L + O(\log^{m-1} L)$ ), mixed exponents in higher dimension, and the weighted primitives for general  $k, h$ .